Nathan Joseph Svoboda

630-532-9604 | njsvoboda04@gmail.com | Tempe, Arizona | ennjaywithagreyhat.github.io/portfolio | github.com/EnnJayWithAGreyHat

Education

Bachelor of Science, Computer Science at *Barrett the Honors College at Arizona State University* *August 2022 - May 2026*

- Awards: Dean's List 5 Semesters

Technical Skills

Programming Languages: Python, JavaScript, C, C++, Java, SQL, Bash
Technical Proficiencies: IDA, BinaryNinja, GDB, Pwntools, x86, Yan85, Wireshark, Nmap, Burpsuite
Project Management: Git, Scrum, ClickUp

Professional Experience

Full-Stack Software Engineering Intern - Hawkeye Energy Solutions *May 2024 - Present*

- Implemented React UI components and state management to enable customers to configure device thresholds, satisfied customer request and improved configuration speed for operators
- Redesigned the backend of a React application to fix API call and data resolution errors by fixing Java methods that would communicate directly with building meters through our integrated servers
- Spearheaded the migration and integration of historical metering data from an older database to a newer and more dynamic database using SQL to provide building metering trends from the past for a client
- Worked in a CI/CD environment for multiple customer-facing applications utilizing npm, Gradle, GitHub, Java, and the Niagara framework to maintain and develop applications and documentation for customers

Software Team Lead - Paragon Autonomous *January 2024 - May 2025*

- Led software development for autonomous fire-surveillance drone and implemented A* pathfinding in C++ using point clouds, enabling autonomous navigation with tests yet to be run on the prototype
- Researched and developed a template framework in C++ to simplify code deployment with flight controllers, and ensured compatibility with Mavros, Gazebo, and Ardupilot for drone programming and functionality
- Tested the Lidar and Ultrasonic sensors to see the information that is provided, and how that data can be integrated into our code for adaptation of flight controls, and collision avoidance

Projects and Involvement

Yellow Belt and Coursework | *GDB, IDA, BinaryNinja, Pwntools* *August 2024 - Present*

- Leveraged pwn.college challenges to develop practical skills in exploiting vulnerabilities and analyzing binaries using IDA, GDB (gef), Python, Bash, and other tools to earn ctf flags and earn my yellow belt
- Crafted and executed buffer overflow, tcache corruption, ROP, Yan85, shellcode injection (with ASLR and NX), DHKE handshakes, SHA collisions, and POA exploits on binaries to simulate attacks on vulnerable programs
- Used example web servers to practice exploits relating to web security, SQL injections, HTTP authentication bypassing, command injection, XSS, CSRF, cookie manipulation, ARP spoofing, and MITM attacks

3rd Place Cactus Con CTF 2026 | *Burpsuite, Python* *February 2026*

- Utilized my skills to earn a flag for the team by finding a web vulnerability related to unsafe image rendering, and utilized a CSRF attack with the Burpsuite repeater to send commands to a remote server
- Learned about the importance of JWT safety, worked on a flag regarding utilizing an error with the JWT RSA signature that allowed our team to use our signature to sign the JWT, get an admin session and a flag
- Expanded my knowledge regarding scripting, privilege escalation, binary exploitation, and reverse engineering techniques through various other challenges

ASU Hacking Club | *GDB, Python, GoLang* *August 2024 - Present*

- Learned how to exploit vulnerabilities in the GoLang game code of Tasteless Shores to capture flags
- Experimented with an outdated CSGO source code, and practiced randomness manipulation along with reversing various binaries, and tracing live game code
- Explored archived CTF challenges, further explored image manipulation with patch directives, sprite modifications, and explored ways to test software security such as fuzzers and dynamic analysis tools